https://github.com/copilot

Gemini3Flash

Q: I am a student who is a first-time learner of Python and its packages. Please provide an at-a-glance summary of how pandas has been used in the attached marimo notebooks (Wk02_Marimo_ZScore.py, Wk03_InteractivePlots_Plotly.py, Wk04_DataPreparation.py) to help me understand its usage better.

Welcome to Python! Think of **pandas** as the "Excel of Python." It is the most important tool for handling tables of data (called **DataFrames**).

Here is an at-a-glance summary of how you used `pandas` across your three notebooks, categorized by the specific tasks you performed.

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1. Data Input & Exploration

In **Wk03** and **Wk04**, you used pandas to bring data into your environment and take a quick look at it.

- `pd.read_csv()`: Loads a `.csv` file into a DataFrame (`df`).
- `df.head()`: Shows just the first 5 rows (great for a quick check).
- `df.unique()`: Lists all distinct values in a column (e.g., finding all unique Sectors in your S&P 500 list).
- `pd.option_context()`: Temporarily changes settings, like allowing you to print *all* rows of a table instead of a truncated version.

2. Selecting & Filtering Data

You used specific pandas techniques to "slice" your data to find exactly what you needed.

- `iloc[:, 0]`: Used in **Wk02** to pick the "Most Recent" column of financial data from Yahoo Finance by its position.
- **Boolean Indexing**: Used in **Wk03** to filter rows, such as `df[df['Sector_Key'] == 'financial-services']`.
- `.get()`: A safe way to grab a value from a column (a **Series**) without the code crashing if that value is missing.

3. Data Cleaning (Handling Missing Values)

Real-world data is messy. You used pandas to fix "holes" in your data (represented as `NaN` or "Not a Number").

- `.isna()` / `.notna()`: Identifies where data is missing or present.
- `.fillna(0)`: Used in **Wk03** to replace missing Debt values with `0`, assuming no report meant no debt.
- `.dropna()`: Used in **Wk04** to remove rows that were missing critical info (like the Sector or Z-Score) before running a regression.

4. Transformation & Variable Construction

This is where you did the "heavy lifting" for your financial analysis in **Wk04**.

- **Vectorization**: Instead of writing a loop to calculate the Z-Score for 500 companies one-by-one, you passed entire columns into your `Zscore()` function. Pandas calculated the whole table at once instantly.
- `.apply()` **with** `lambda`: Used to categorize data. For example, you turned a numerical Z-Score into a text label like "Safe Zone" or "Distress Zone."
- `.sort_values()`: Organized your data by Ticker and Year so that chronological operations would work correctly.
- `.groupby()` **and** `.shift(1)`: **(Crucial for Finance)** You used this to create "Lagged" variables. It tells pandas: *"For each company, look at the row above it (the previous year) and pull that value into the current year."* This allowed you to compare *last year's* risk to *this year's* cost of debt.

5. Data Analysis & Output

Finally, you used pandas to summarize your findings and save your work.

- `pd.crosstab()`: Created a "Contingency Table" (like an Excel Pivot Table) to show how many companies fell into "High Risk/High Cost" vs. "Low Risk/Low Cost" categories.
- `df.to_csv()`: Exported your cleaned, calculated data to a new file so you wouldn't have to re-run the 30-minute download process again.

Summary Table for Your Reference

Function	What it does	Example from your code
DataFrame	The main table object	`df = pd.read_csv(...)`
Series	A single column	`df['Total_Debt']`
`.iloc`	Select by position	`balance_sheet.iloc[:, 0]`
`.shift()`	Gets previous row	`df.groupby('Ticker')['Z_Score'].shift(1)`
`.fillna()`	Fills missing values	`df['Total_Debt'].fillna(0)`
`.apply()`	Runs a function on every row	`df['Z_Score'].apply(categorize_zscore)`
`.crosstab()`	Frequency table	`pd.crosstab(index=rows, columns=cols)`

Pro Tip: As you continue, remember that almost any time you think "I need to do a calculation on this table," there is likely a `df.something()` command in pandas that can do it without you needing to write a manual `for` loop!